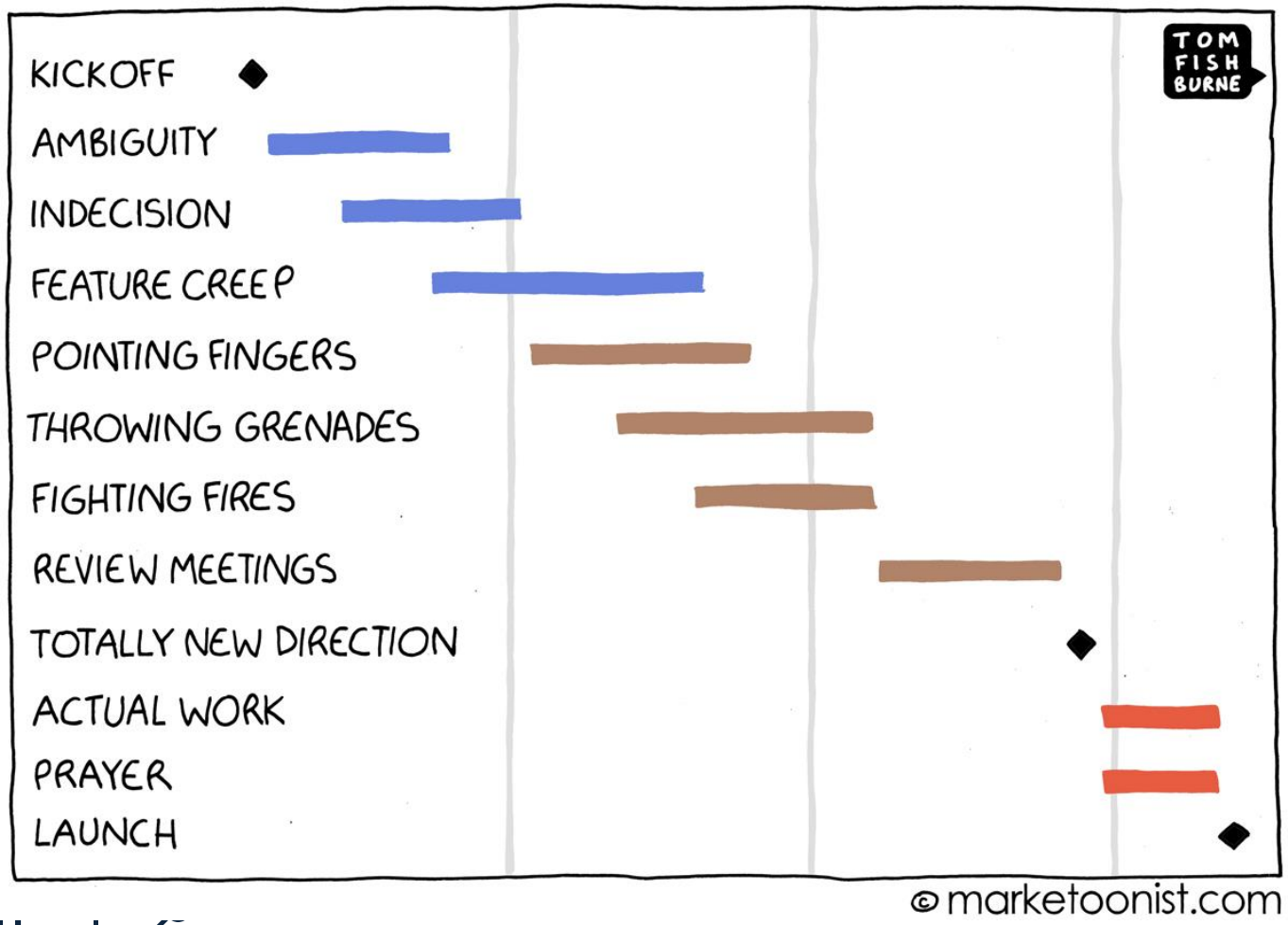


ASD03 Project Management 2

Johan Bontes (johan@digitsolutions.nl)

CSC3003S

30 JULY 2025

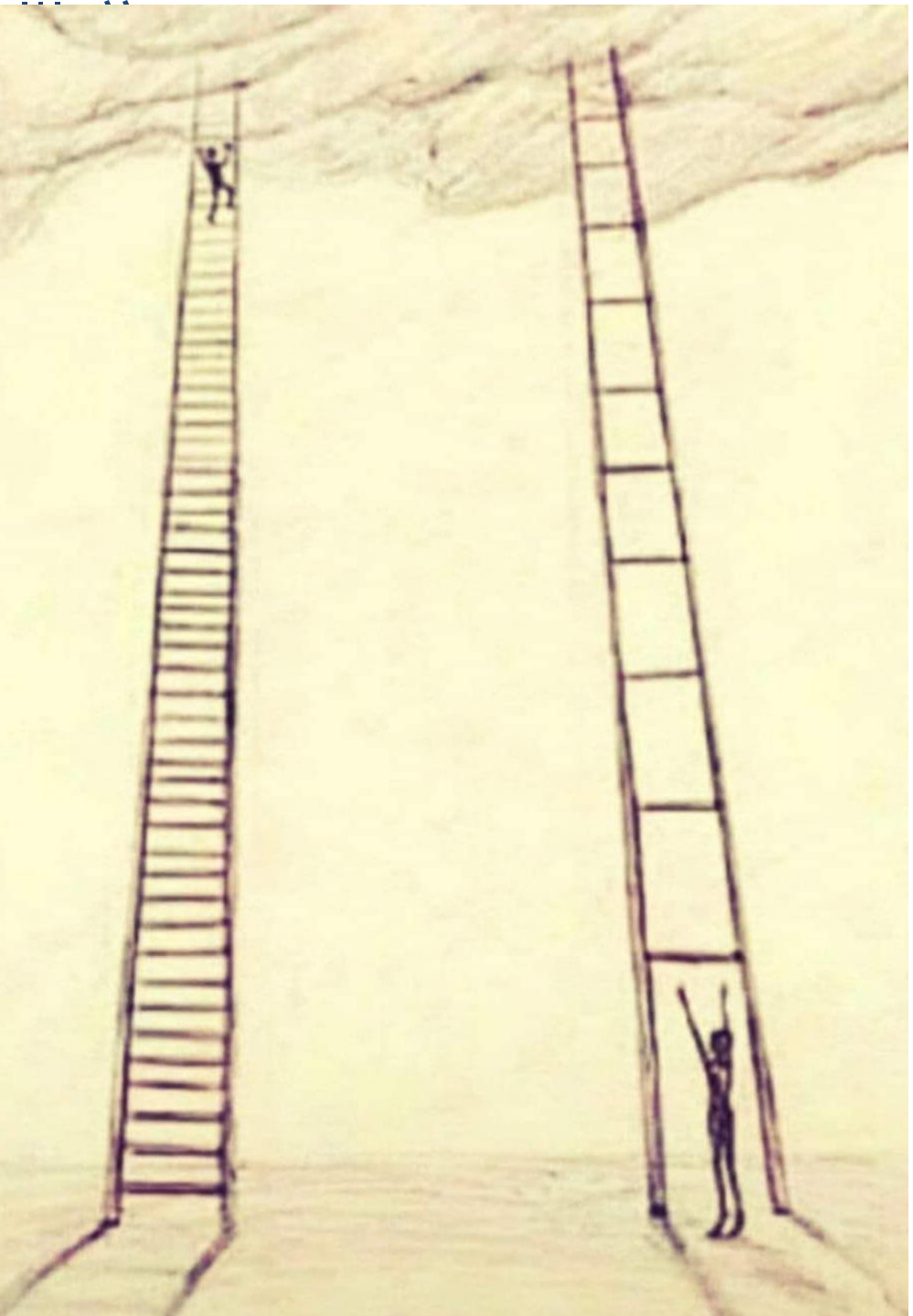


CSC3003S ASD
Project Management 2

A. Scheduling

B. Network Analysis

C. Risk

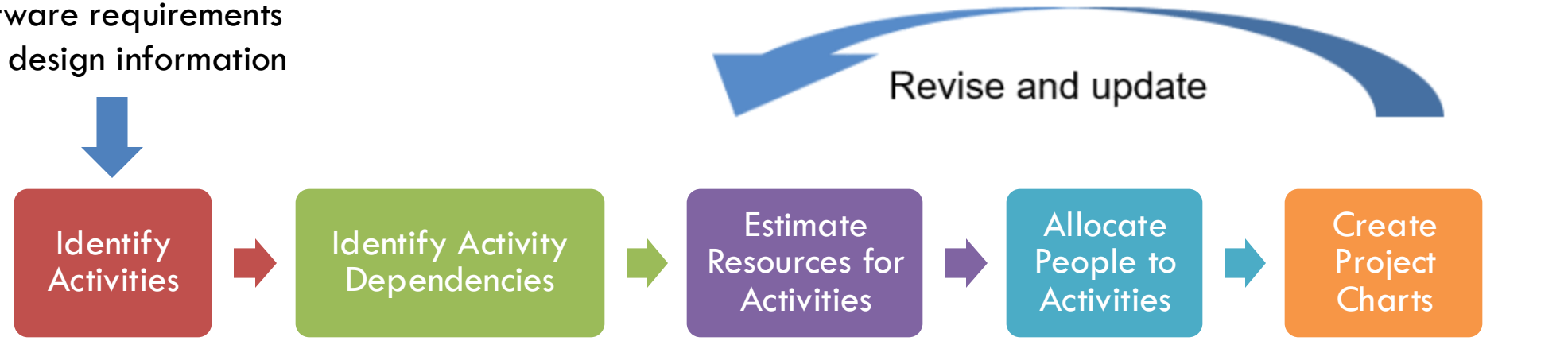


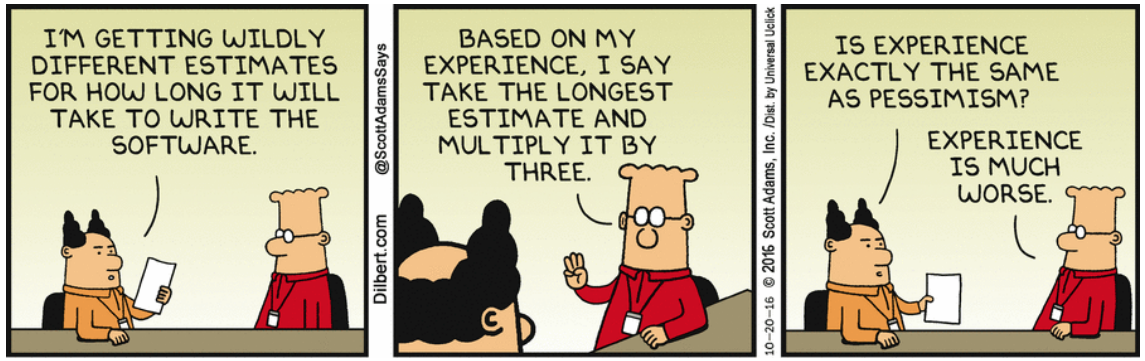
Project Scheduling

- Split the work in a project into separate tasks
  - **Minimize task dependencies**
- Assign the work to team members
- Estimate time needed to complete each task
  - Split up if much longer than 1 week (never over 2 months)
  - Make tasks concurrent to make optimal use of workforce.
- Estimate the effort required
  - Who will work on the tasks?
  - Resources needed to complete each task
- Mostly **needs project manager's intuition and experience**

Project Scheduling Process

Software requirements and design information





Gantt charts describing the project schedule

non-agile, from Figure 23.4 in Sommerville

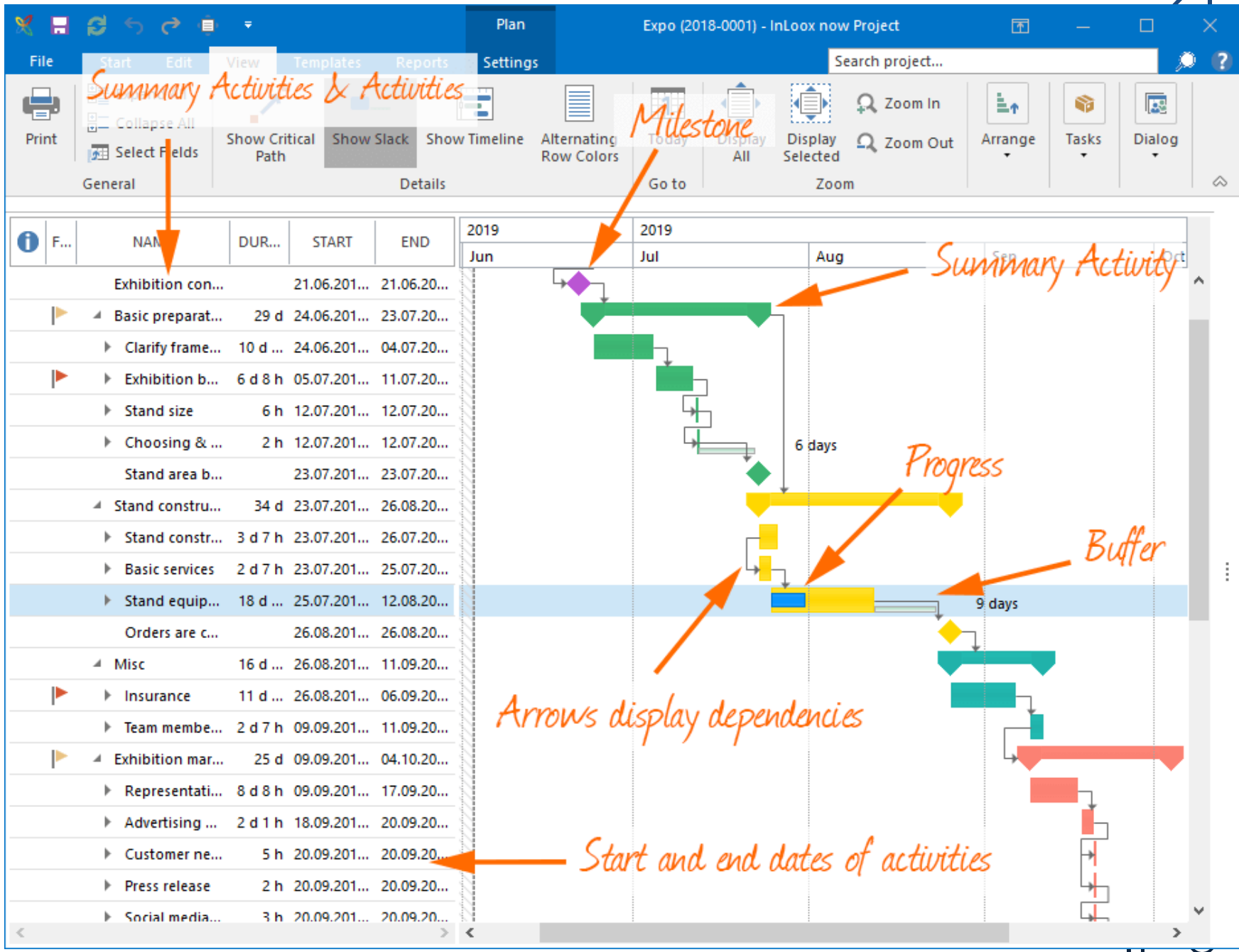
Gantt Charts

- Horizontal axis: time
- Vertical axis: project tasks

Capture:

- Task completion
- Simple dependencies
- Milestones and deliverables

Can't handle complex task dependencies



Before Project

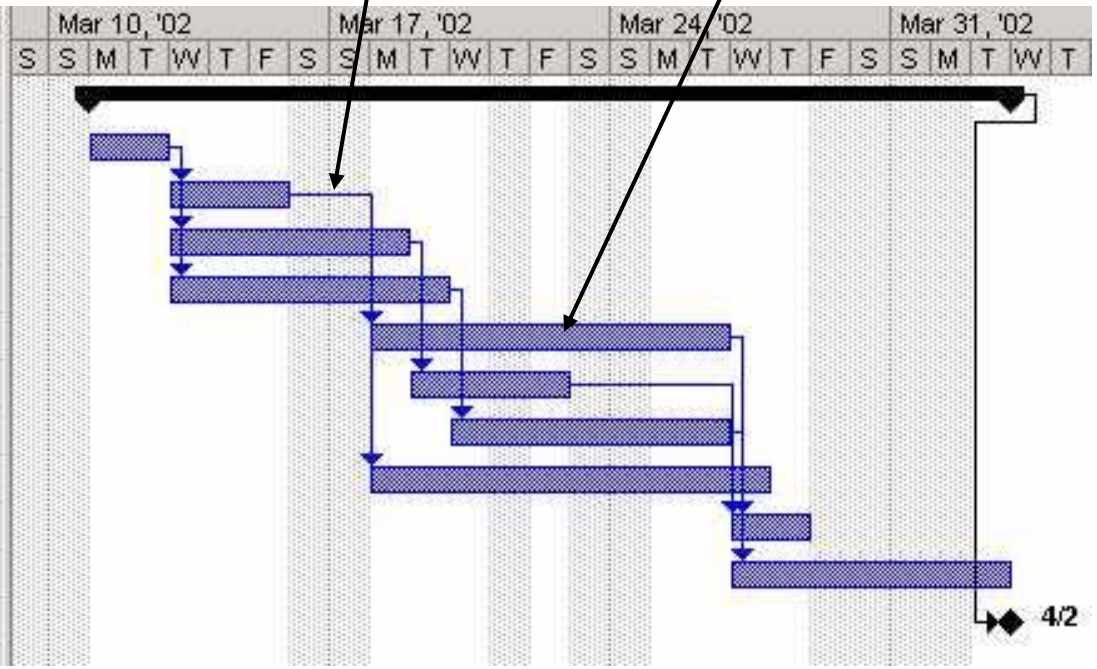
The timeline provides a chronological reference

Activity bars represent activity duration - the longer the bar, the longer the duration

Major project deliverables listed in bold and capped with a diamond

Arrows represent dependencies

	Task Nam	Duration	Start	Finish	Prede
1	**Proj**	**14 days**	**Mon 3/11/02**	**Tue 4/2/02**	
2	A	2 days	Mon 3/11/02	Tue 3/12/02	
3	B	3 days	Wed 3/13/02	Fri 3/15/02	2
4	C	4 days	Wed 3/13/02	Mon 3/18/02	2
5	D	5 days	Wed 3/13/02	Tue 3/19/02	2
6	E	6 days	Mon 3/18/02	Tue 3/26/02	3
7	F	3 days	Tue 3/19/02	Fri 3/22/02	4
8	G	4 days	Wed 3/20/02	Tue 3/26/02	5
9	H	7 days	Mon 3/18/02	Wed 3/27/02	3
10	I	2 days	Wed 3/27/02	Thu 3/28/02	6,7
11	J	3 days	Wed 3/27/02	Tue 4/2/02	8
12	Comple	0 days	Tue 4/2/02	Tue 4/2/02	1



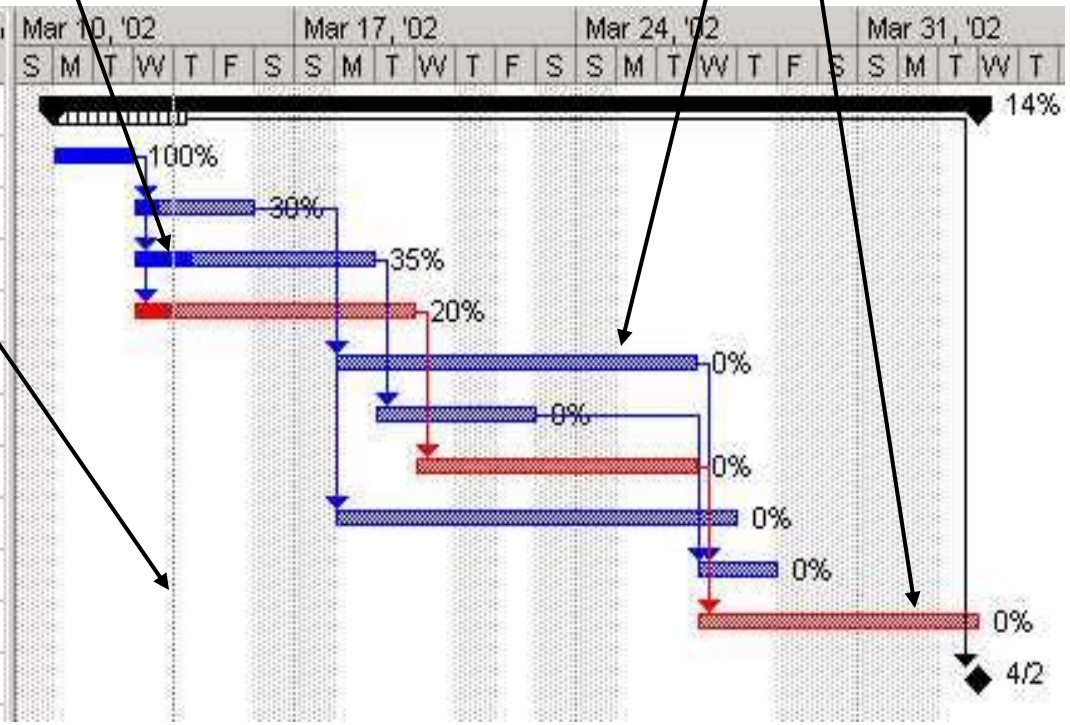
During Project

Current date represented by a dotted vertical line

Colour-coding assigns tasks to team members

Bars filled to show progress

	Task Narr	Duration	Start	Finish	Predi
1	Proj	14 days	Mon 3/11/02	Tue 4/2/02	
2	A	2 days	Mon 3/11/02	Tue 3/12/02	
3	B	3 days	Wed 3/13/02	Fri 3/15/02	2
4	C	4 days	Wed 3/13/02	Mon 3/18/02	2
5	D	5 days	Wed 3/13/02	Tue 3/19/02	2
6	E	6 days	Mon 3/18/02	Tue 3/26/02	3
7	F	3 days	Tue 3/19/02	Fri 3/22/02	4
8	G	4 days	Wed 3/20/02	Tue 3/26/02	5
9	H	7 days	Mon 3/18/02	Wed 3/27/02	3
10	I	2 days	Wed 3/27/02	Thu 3/28/02	6,7
11	J	3 days	Wed 3/27/02	Tue 4/2/02	8
12	Comple	0 days	Tue 4/2/02	Tue 4/2/02	1



Kanban Boards

Visualize current project activities at a glance

Each task is represented as a single card, usually a user story.

Cards move from left => right as they are completed:

1. Stories/Backlog: The full list of tasks
2. To do: Tasks planned for current cycle or phase
3. In progress: Tasks that have been started
4. Testing: Tasks that have been completed and are ready to undergo testing
5. Done: Tasks that are completed

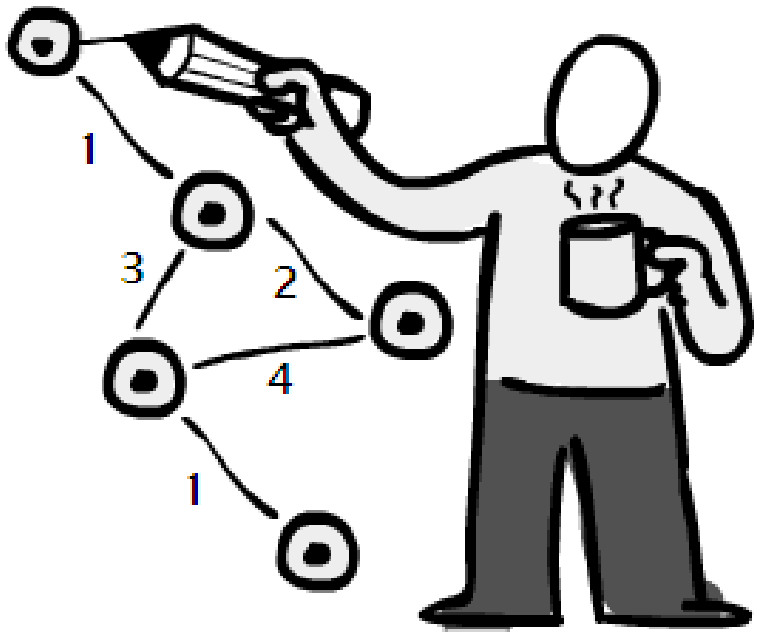


Kanban Task Management



Backlog → Specs Ready → Development → Testing → Deployment → Done

New → To Do → Doing → Testing → Done



CSC3003S ASD03
Project Management 1

A. Scheduling

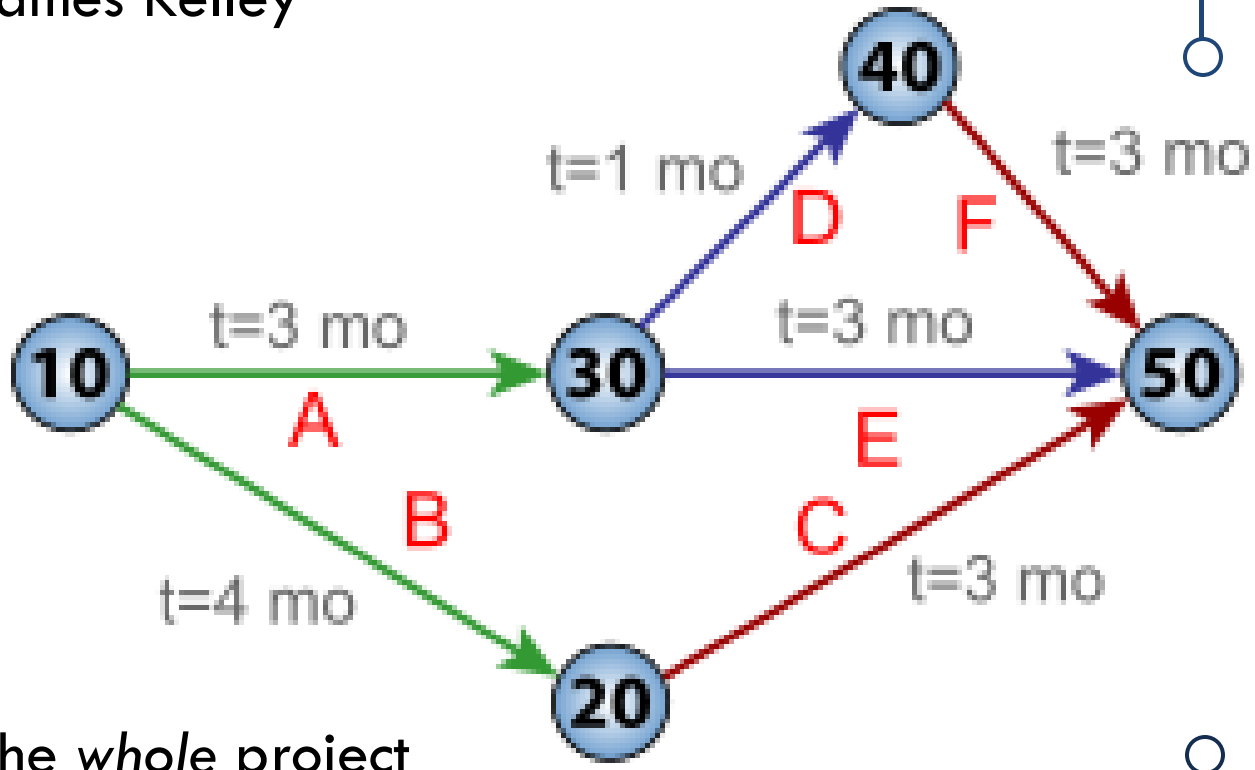
B. Network Analysis

C. Risk

Critical Path or Network Analysis

Developed in the 1950s by Morgan Walker & James Kelley

Critical path
= *The longest stretch of dependent activities*
and the total time required to complete them
= *shortest time project can be done (here: A-D-F)*



- Visualised in PERT charts (outdated)
  (Program Evaluation Review Technique)
- Any delay of a task on a critical path delays the *whole* project

Tasks for a Multiuser 3D Meeting Place

#	Activity	Description	Length	Depend
1	Access	Login: username + password and access privileges to system.		
1.1	User's status	keep a database of users status (online, available, busy, etc.).	30	
1.2	Persistence	to keep a database of all avatars, objects, their positions and states; thus providing "persistence" in the virtual world for users who come and leave.	180	1.1
1.3	Inventory	keep a database of users virtual inventory and support the addition, removal and exchange of items in the inventory.	10	1.2
2	Logging	log all events in world		
2.1	Log user events	log user status in world	10	1.3
2.2	Log avatar events	log position of avatars	10	3.4
2.3	Log object events	log position of objects.	10	4.3
2.4	Log communication	log all communication that occurs in VE.	100	2.1,2.2,2.3

What is Network Analysis?

- Project tasks (activities):
  - Are often interdependent
  - But need to be done *in parallel* for teamwork to be effective
- Task networks are graphical depictions of task dependence
- Network analysis is a project planning method that:
  - Determines the critical path
  - Establishes “most likely” time estimates
  - Calculates boundaries to stop project slippage

Terminology

Earliest Start/Finish

- Earliest a task can begin/end if *all preceding tasks are completed* in the shortest time

Latest Start/Finish

- Latest a task can begin/end *without delaying* the minimum project completion time

Critical Path

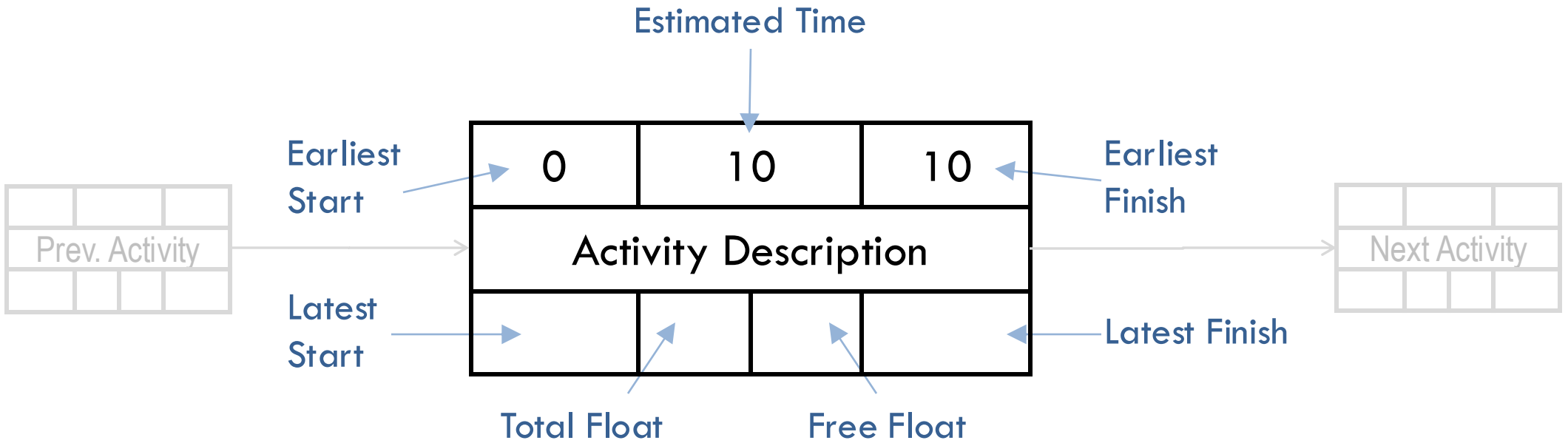
- Chain that determines **overall project duration**
- Multiple paths can be critical

Slack (Float)

- The amount of leeway allowed on a task whilst still maintaining the critical path

Network Analysis

Activity-on-node diagram (replaces PERT)

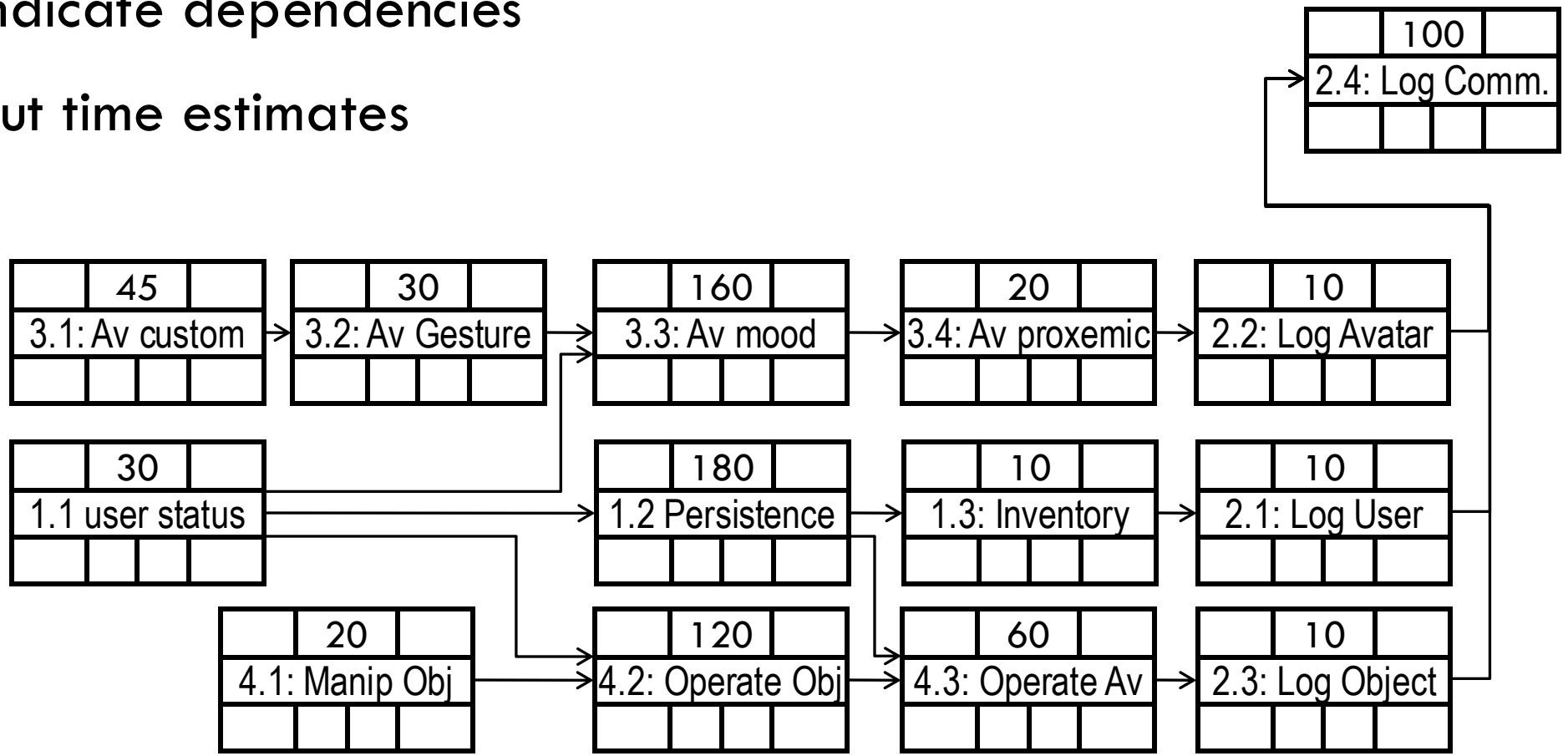


Tasks are shown as boxes

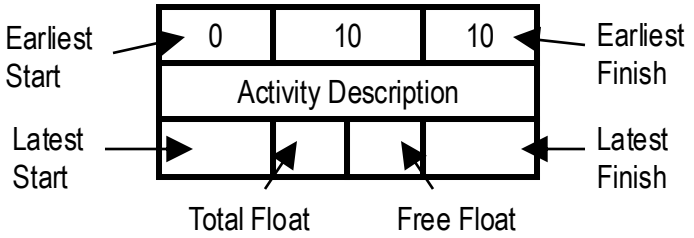
Sequence constraints are lines connecting the boxes

Analysis: Step #1

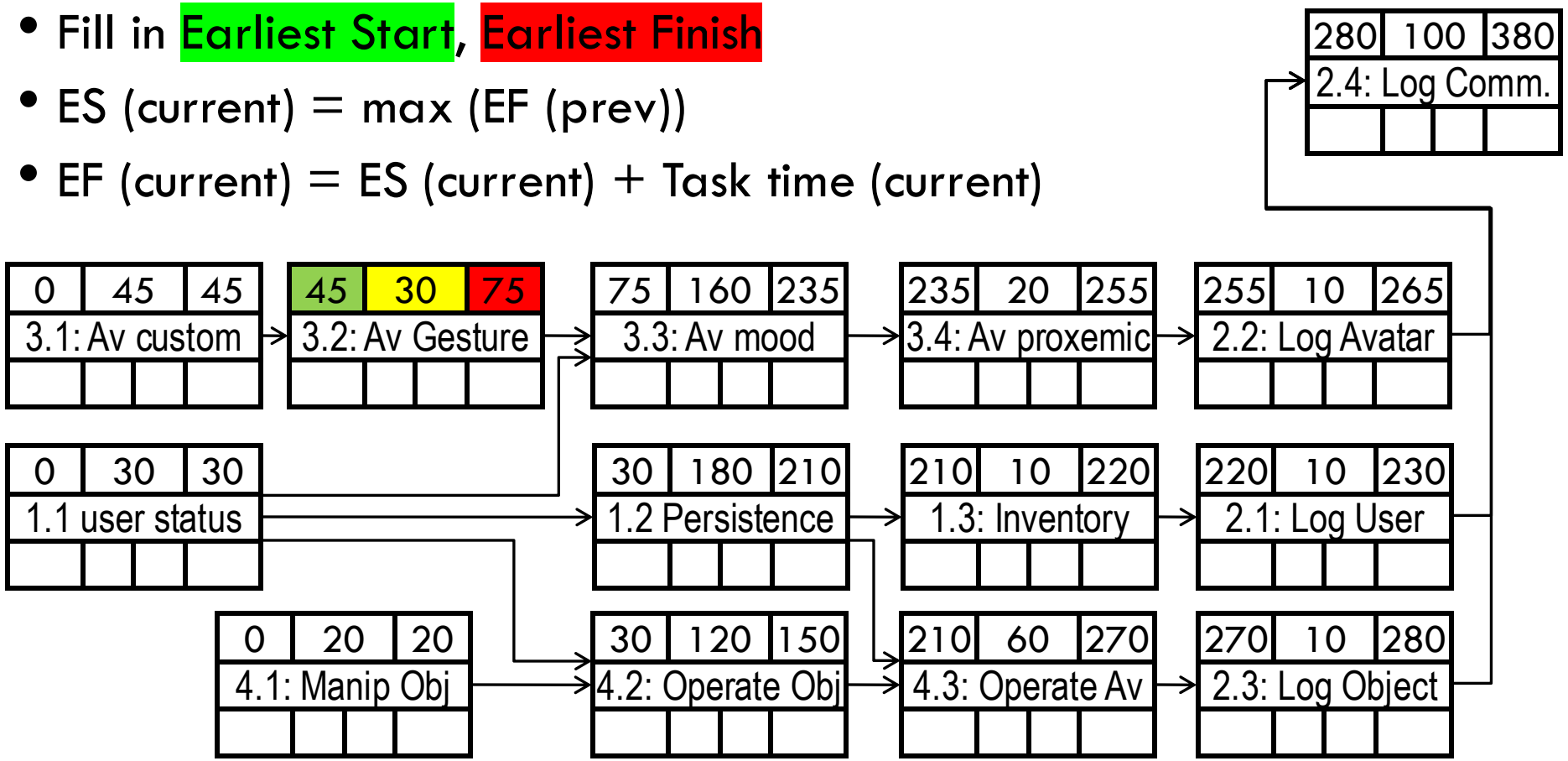
- Label tasks in order
- indicate dependencies
- Put time estimates



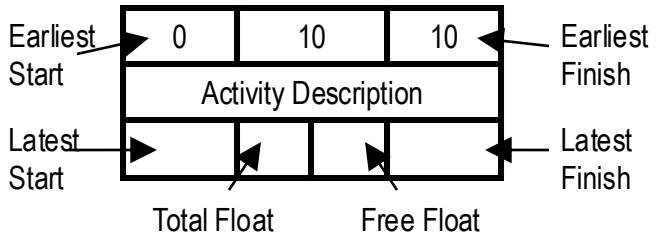
Analysis: Step #2



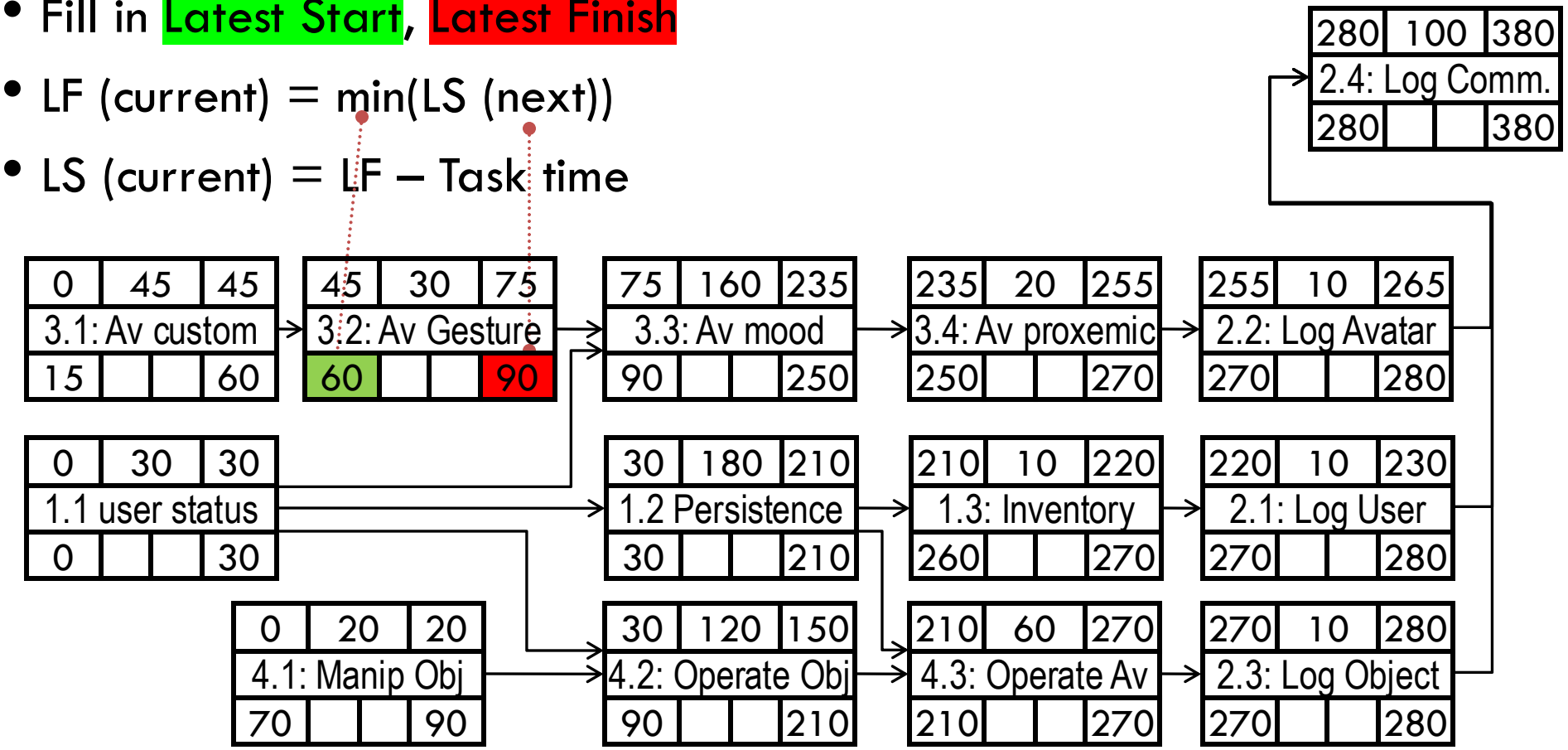
- From the estimated time for each activity
- Carry out a "forward pass"
- Fill in Earliest Start, Earliest Finish
- ES (current) = max (EF (prev))
- EF (current) = ES (current) + Task time (current)



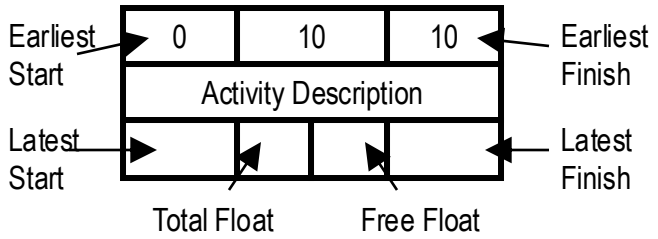
Analysis: Step #3



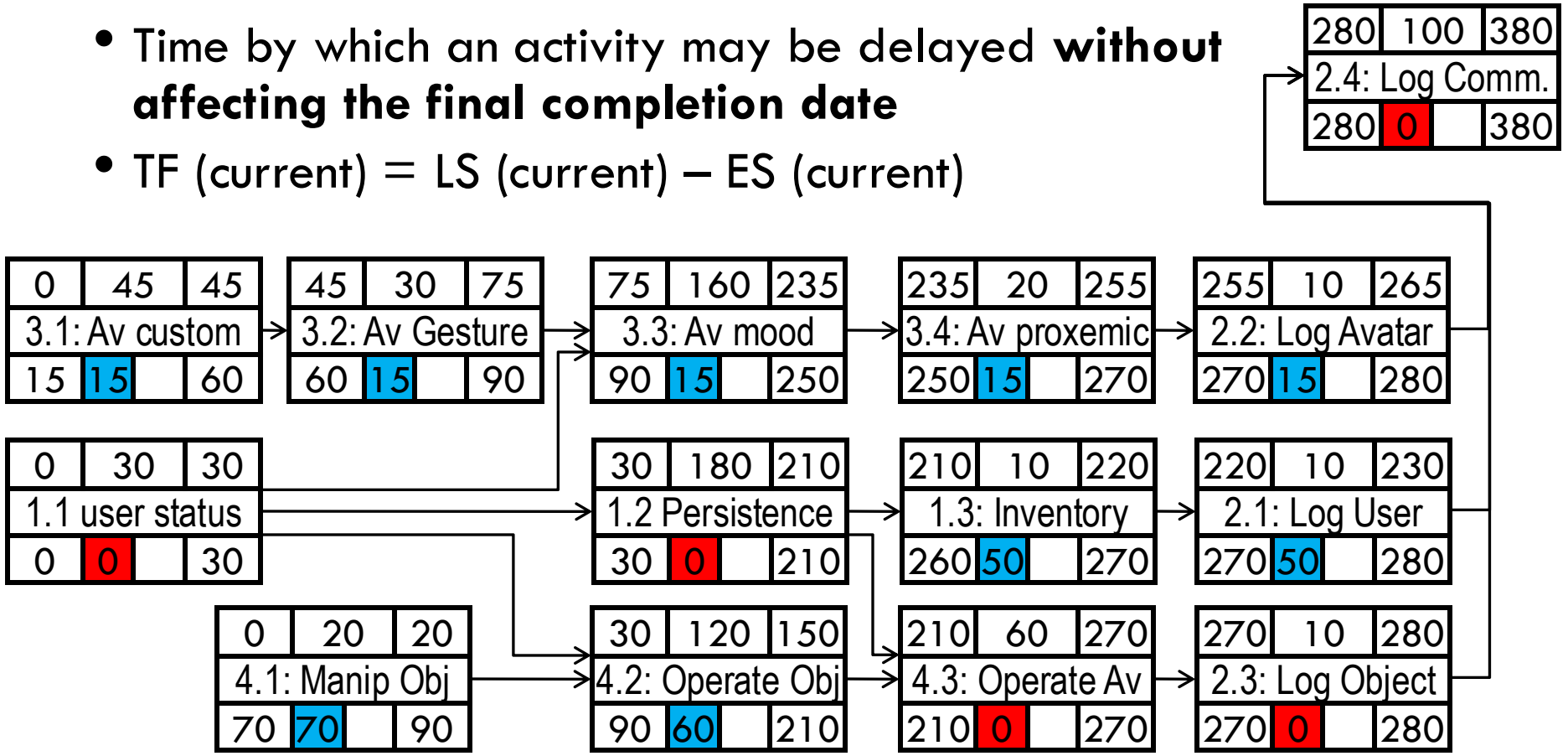
- Carry out a "backward pass"
- Fill in Latest Start, Latest Finish
- LF (current) = min(LS (next))
- LS (current) = LF – Task time



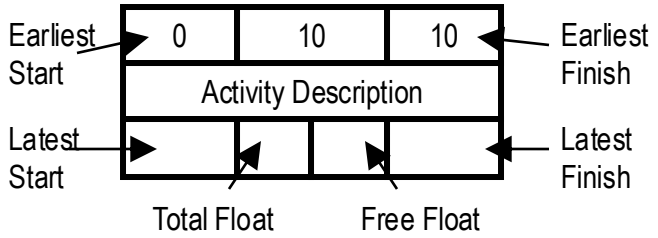
Analysis: Step #4



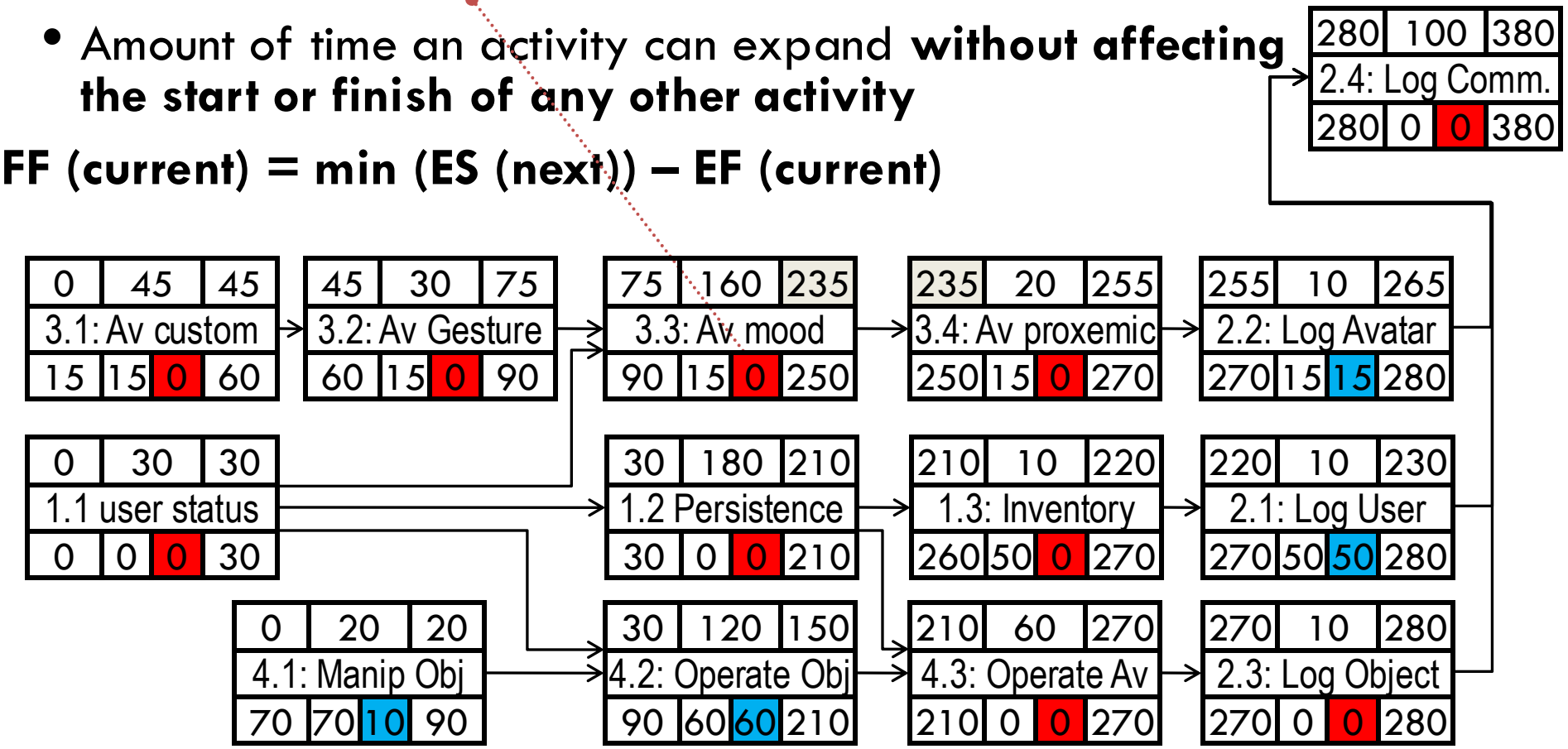
- Fill in Total Float
  - Time by which an activity may be delayed **without affecting the final completion date**
  - TF (current) = LS (current) – ES (current)



Analysis: Step #5

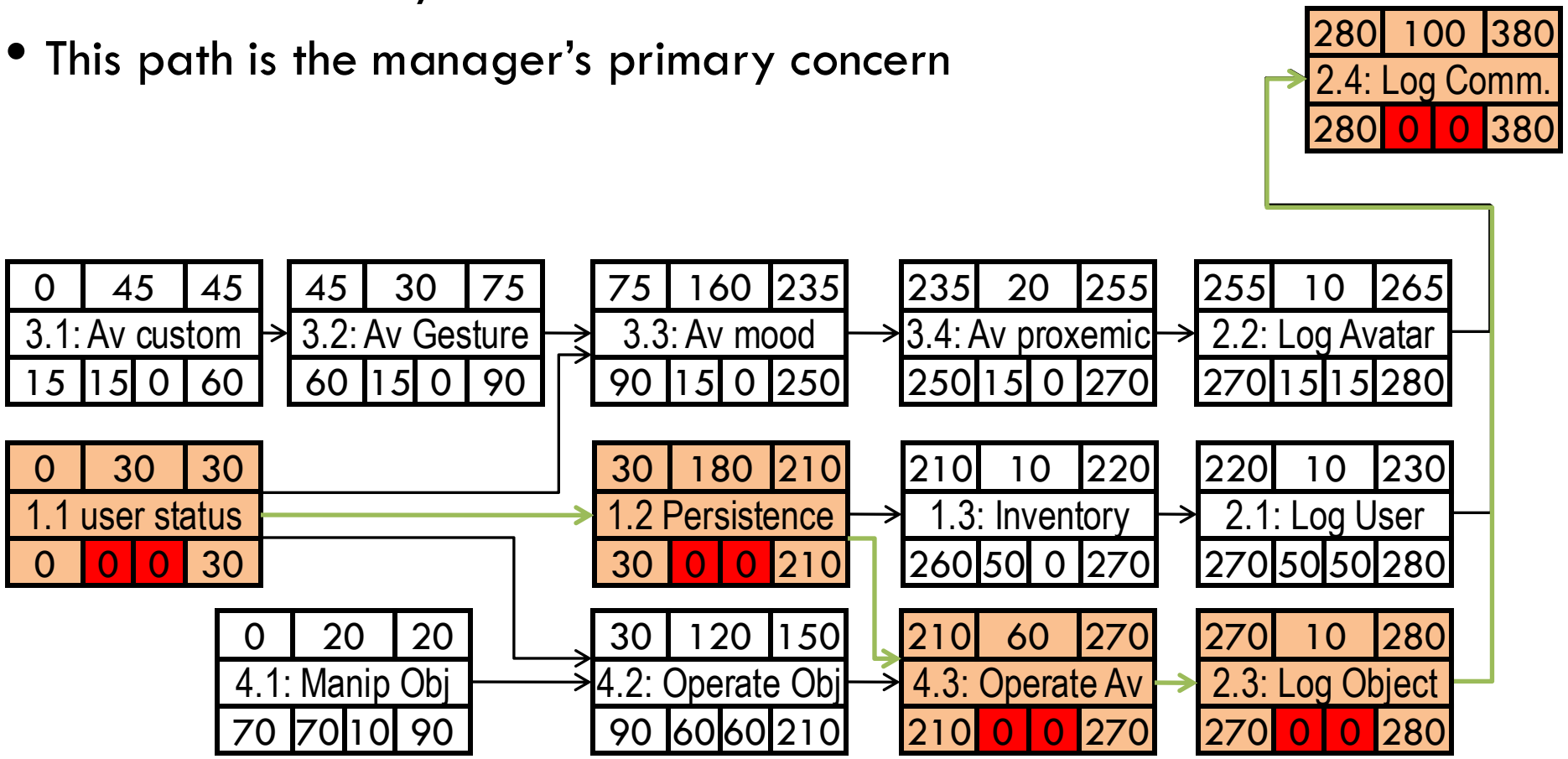


- Fill in Free Float
  - Amount of time an activity can expand **without affecting the start or finish of any other activity**
- **FF (current) = min (ES (next)) – EF (current)**



Analysis: Step #6

- Critical Path
  - Does not have any float
  - This path is the manager's primary concern



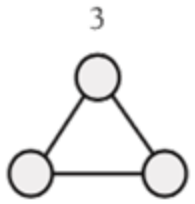
Problems with CPM

- It assumes task time is exact
  - May cause near-critical paths to be deprioritised
- Does not take into account human factors (~30% of delays)
  - Task-switching: causes delays due to time needed to get back into 'flow'
  - Student syndrome: delay work on task until the last minute
  - Parkinson's Law [1]:

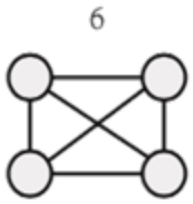
    - work expands so as to fill the time available for its completion
  - Overhead due to communication: $\frac{1}{2}n(n-1)$



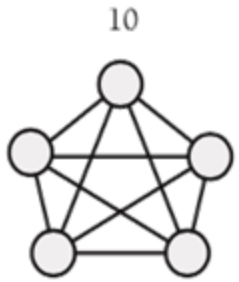
Communication path with two programmers



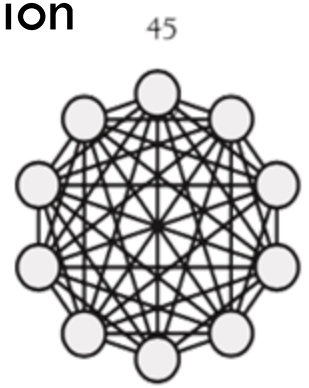
Communication paths with three programmers



Communication paths with four programmers



Communication paths with five programmers



Communication paths with ten programmers

[1] C. Parkinson, *Parkinson's Law*, The Economist. London, 1955

CCPM: Critical chain project management

CCPM = CPM +

- Resource dependencies: add to to obtain non-human resources
- Don't optimise for single solution
  - Inherent uncertainty in task time > Δ optimal/near optimal solutions
- Identify and insert buffers
  - Project buffer: (one per project) extra time at the end of project
  - Feeding buffer (extra time on non-critical-chain tasks)
  - Resource buffer (money, etc. allocated to complete project parts)
- Monitor consumption of buffers, instead of fixating on critical path

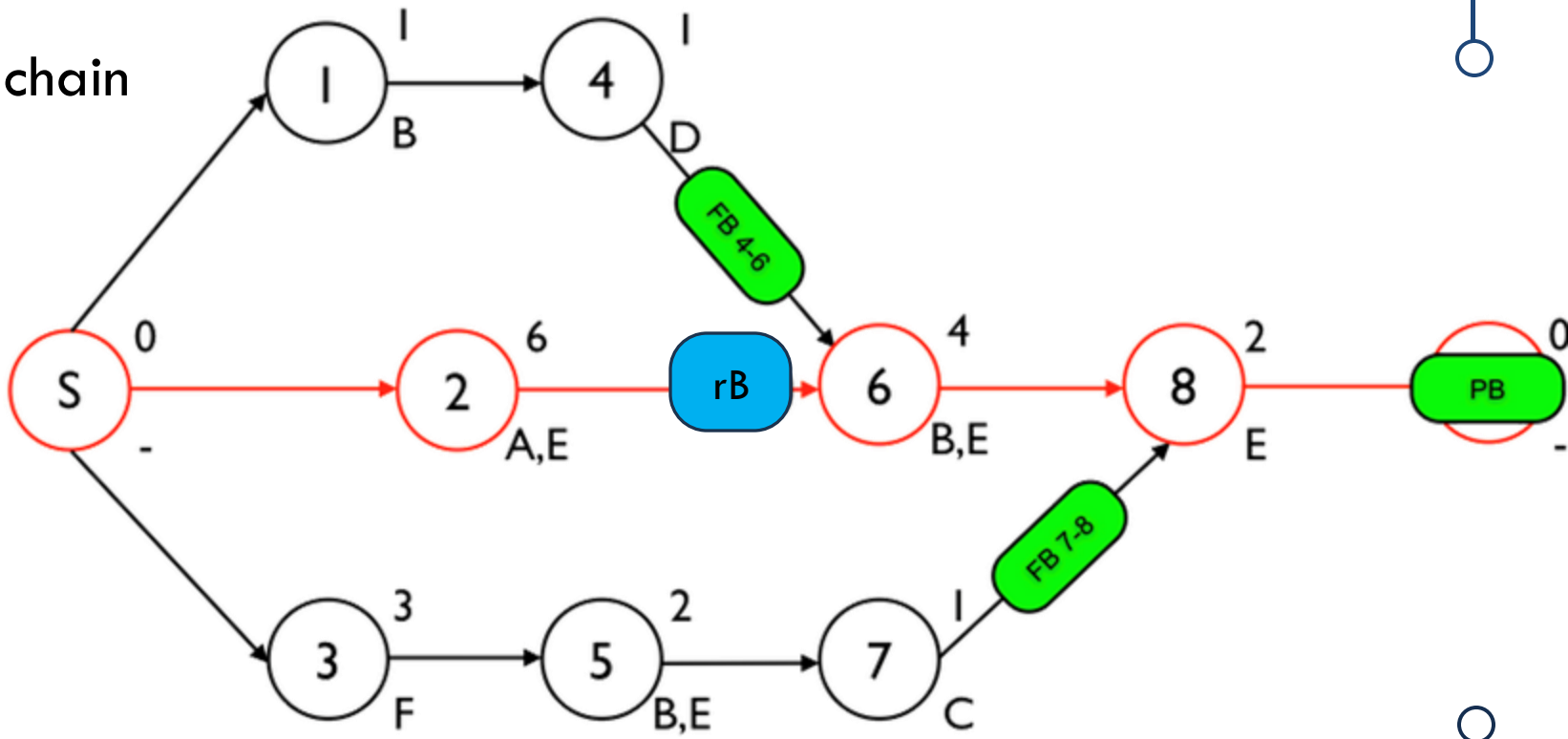
Feeding / resource buffer

The feeding chain:

- All paths '*feeding*' into the critical chain

Feeding buffer:

- protects critical chain (CC) from delays in feeding chain
- directly in front of CC entrance



Resource buffer:

- warning that resource is needed at CC stage
- put resources into buffer before needed, take out of buffer when needed

CCPM planning

1. Identify tasks, dependencies, and time, same as CPM
2. Split time into optimistic A and pessimistic B estimates
   A estimates can be achieved with 50% probability (maybe faster/slower)
   B estimates with 95% probability
3. max(time[B]) – max(time[A]) => buffers (project/feed/resource)
4. Assign resources to each task
   I. schedule: use time A to **resource level** each task
   II. longest sequence of tasks = **critical chain**
   III. time overruns on critical chain: buffers– underruns: buffers+
5. Execution
   I. Planned schedule **never changes**, actual schedule tracks new critical path
   II. hand over tasks ASAP (*relay race: reduces procrastination*)
6. Monitor
   I. Track buffer status: buffers full = project on target

Alternatively/additionally use Monte Carlo simulation

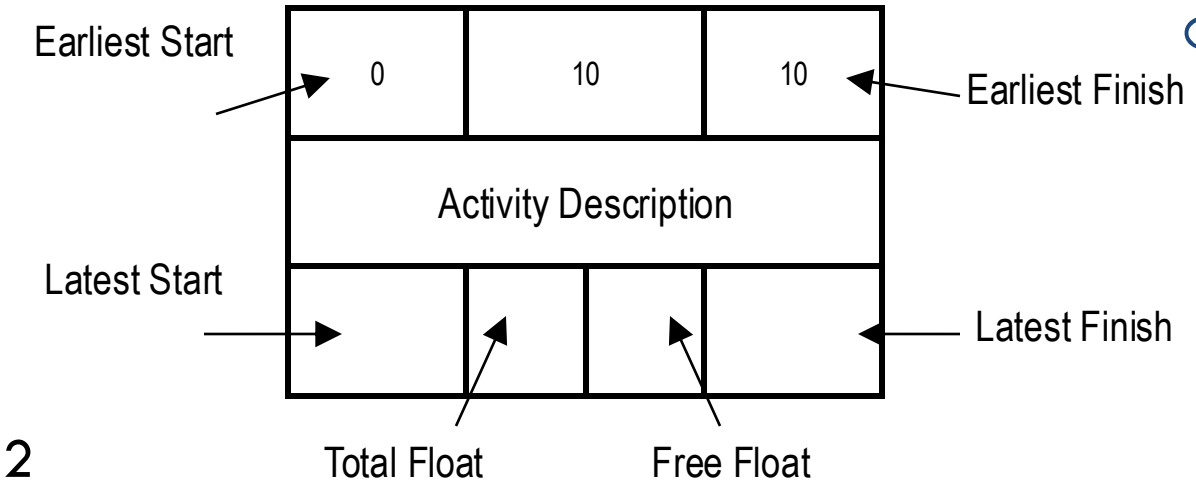
https://openlibrary.org/books/OL51882M/Critical_chain_project_management

Exercise: Network Analysis

- Do network analysis for a small development project

Task	Time	Dependencies
A: Analysis	6	
M: Modelling	8	A
I1: Implementation (Phase 1)	12	A
I2: Implementation (Phase 2)	8	I1
I3: Implementation (Phase 3)	14	M
UX: User Testing	5	I2, I3
D: Documentation	10	A
I: Installation and Customer Hand Over	10	D, UX

Solution: Do it first!

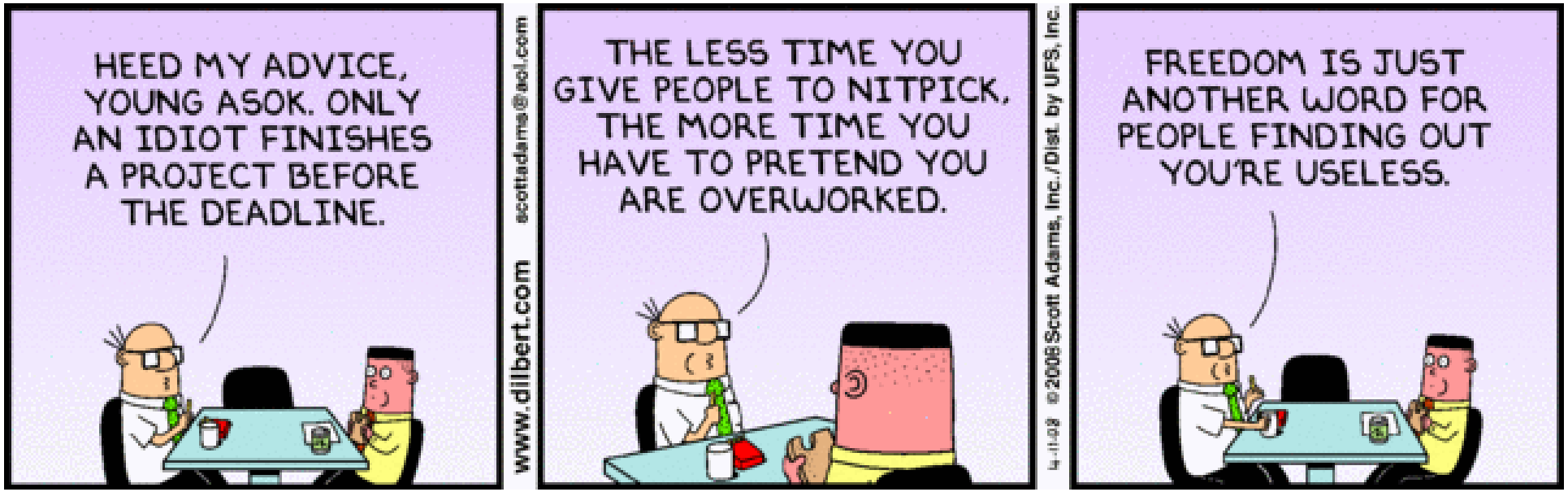


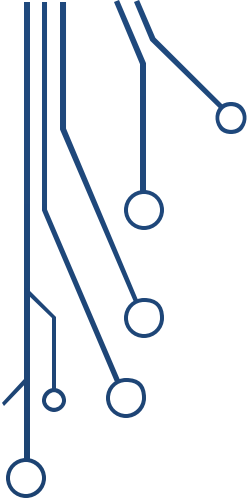
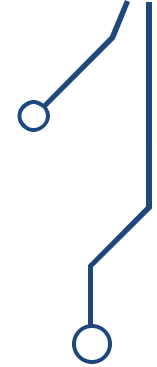
See ASD-02-X-NetAnalyseEx.pdf

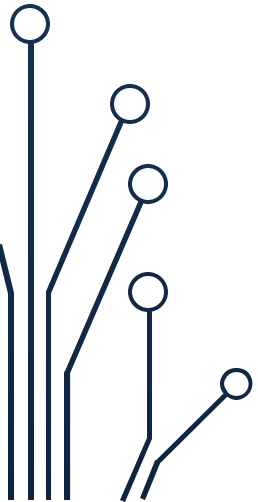
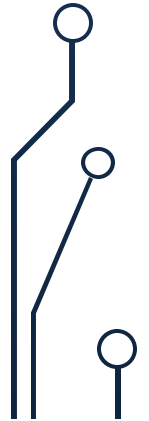
Password:

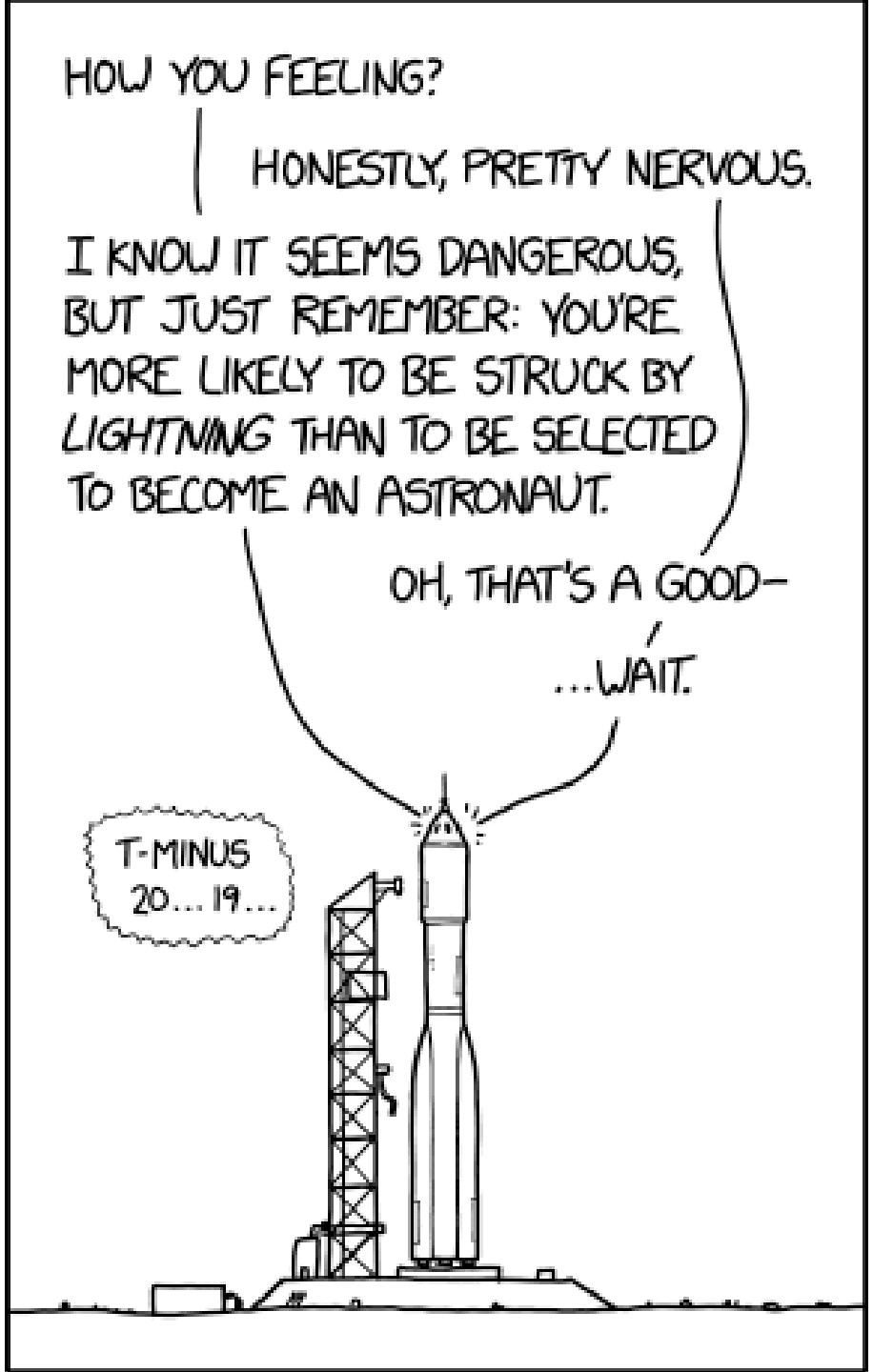
bottom row of activity I2: Implement 2

(as a six digit number)





CSC3003S ASD
Project Management

A. Scheduling

B. Network Analysis

C. Risk

Managing Risks

- Why?
  - Projects have a *high level of uncertainty*
  - Better to *anticipate* problems
- How?
  - Identify → Analyze → Rank
  - Plan for monitoring, mitigation, management
  - Revisit continually during project



Risk

Risk = probability × cost

Cost → Probability ↓	Cheap ☐	Expensive 
Unlikely 😐	Accept	Transfer (insure)
Likely 😨	Mitigate	Avoid

Boehm's top ten risk items + 2

1 Personnel shortfalls, failure to recruit or retain key staff	2 Unrealistic schedules and budgets	3 Developing the wrong functions	4 Developing the wrong user interfaces	5 Gold-plating
6 Continuing stream of requirements changes	7 Shortfalls in externally-performed tasks	8 Shortfalls in externally-furnished components	9 Real-time performance shortfalls	10 Straining computer science capabilities
* Being blindsided by the competition	* Outside interruptions (e.g., ill-health)			

Risk Consequences

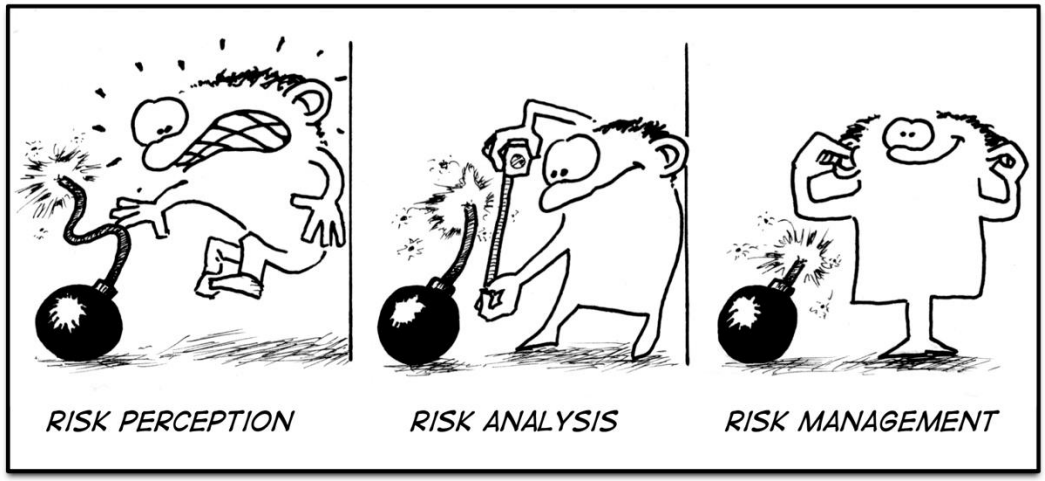
Assess the impact of potential consequences of the risk to:

- Project Timelines
- Project Cost
- Project Feasibility
- Project Quality

	Consequence				
People	Minor skills impact.	Minor impact to capability	Unavailability of core skills affecting services.	Unavailability of critical skills or personnel	Protracted unavailability of critical skills/people.
	Minor injury or first aid treatment	Injury requiring treatment by medical practitioner	Major injury / hospitalization	Single death and/or multiple major injuries	Multiple deaths
Information	Compromise of information otherwise available in the public domain.	Minor compromise of information sensitive to internal or sub-unit interests.	Compromise of information sensitive to this organisation operations.	Compromise of information sensitive to organisational interests.	Compromise of information with significant ongoing impact.
Property & Equipment	Minor damage or vandalism to asset.	Minor damage or loss of <5% of total assets	Damage or loss of <20% of total assets	Extensive damage or loss <50% of total assets	Destruction or complete loss of >50% of assets
Reputation	Local mention only. Quickly forgotten. Freedom to operate unaffected. Self-improvement review required	Scrutiny by Executive, internal committees or internal audit to prevent escalation Short term local media concern. Some impact on local level activities	Persistent national concern. Scrutiny required by external agencies. Long term 'brand' impact.	Persistent intense national public, political and media scrutiny. Long term 'brand' impact. Major operations severely restricted.	International concern, Governmental Inquiry or sustained adverse national/international media. 'Brand' significantly affects organisational abilities.
Financial	1% of Project or Organisational Annual Budget	2-5% of Project or Organisational Annual Budget	5-10 % of Project or Organisational Annual Budget	> 10% Project or Organisational Annual Budget	> 30% of Project or Organisational Annual Budget
Capability	Minimal impact on non-core business operations. The impact can be dealt with by routine operations.	Some impact on business areas in terms of delays, systems quality but able to be dealt with at operational level	Impact on the organisation resulting in reduced performance such that targets are not met. Organisations existence is not threatened, but could be subject to significant review or changed ways of	Breakdown of key activities leading to reduction in performance (eg. service delays, revenue loss, client dissatisfaction, legislative breaches). Survival of the project/activity/organisation is threatened	Critical failure(s) preventing core activities from being performed. The impact threatens the survival of the project or the organisation itself.
	Insignificant	**Negligible**	**Moderate**	**Extensive**	**Significant**

Risk steps

- **Perception**
  - Identify risks
- **Analysis**
  - Classify: kind, cost, likelihood
  - Define metrics to track risks
- **Strategy:**
  - LX: **avoid!** *change requirements*
  - UC: accept
  - UX: transfer *e.g. buy insurance*
  - LC: mitigate (try to reduce likelihood/impact)
- **Monitoring:**
  - Track risk metrics
- **Management:**
  - Execute risk strategy



Cost → / Probability ↓	Cheap	Expensive
Unlikely	Accept	Transfer (insure)
Likely	Mitigate	Avoid

Examples of Risk I

Risk Condition	Consequence	Cat	Prob	Impact	Strategy	Monitoring	Management
Competitors duplicate technology rapidly	Competitors products dilute market share	Follow-On	High	Critical	Develop Copy and IP protection, protect trade secrets through NDAs and source code through copyright	Keep abreast of current research and potential competitors	Adjust product pricing, develop upgrades
Product not well targeted to market	Unable to Capture Market Share	Commercialization	Medium	Critical	Create Business Plan and adapt development accordingly	Tracking current and predicting future market trends	Further market research and retargeting of product

Examples of Risk II

Risk Condition	Consequence	Cat	Prob	Imp-act	Mitigation	Monitoring	Management
Failure to recruit suitable lead developer on schedule	Significant delays (lead developer is on critical path)	Development	Medium	Critical	Advertise widely, offer attractive salary package	Track number and quality of applicants	Second round of recruiting (perhaps internationally), Target specific individual
Failure to acquire international business partner	International marketing and distribution hampered	Follow-On	Medium	Critical	Begin search early, develop business plan and prototypes to promote concept	Keep record of potential partners and commitment status	Delay follow-on until partner recruited or revise business strategy

Risk: Competition in Technology

Risk Condition	Consequence	Strategy	Monitoring	Management
Competitors duplicate technology rapidly	Competitors products dilute market share	Develop Copy and IP protection, protect trade secrets through NDAs and source code through copyright	Keep abreast of current research and potential competitors	Adjust product pricing, develop upgrades



Update 9/5/20..	Update 23/10/20..
Virtools and Prof. Kaufman identified as key competitors	3D Games Studio is a competitor, but not Prof. Kaufman

B. RISKS AND RISK MANAGEMENT/MITIGATION

Risk Number	Risk Description	Impact (1 – 4)	Probability (1 – 5)	Management/Mitigation
1.	Distributed Surveys are not responded to by enough participants.	3	4	Mitigation: 1. The project is structured in such a manner as to be viable with absolutely no survey information. Management: 1. The project proceeds without survey information.
2.	We are unable to secure/maintain a suitable licence for a cloud-based web services provider, either dure to a funding issue	3	2	Mitigation: 1. Prior to development, we will explore multiple solutions and subscription plans in this space. If no suitable licence is found, we will settle on a non-cloud based, local solution that isn't tied to a subscription. Management: 1. We will immiediately shift to a non-cloud based, local solution that isn't tied to a subscription.
3.	We are unable to develop a solution that satisfies *Milk Matters*.	4	1	Mitigation: 1. We work with *Milk Matters* to determine project requirements that are satisfactory to them.
				Management: 1. Parts of the project that do not satisfy them are not deployed, even if this includes the entire project.
4.	The project's scope develops out of our reach and cannot be developed in time.	4	2	Mitigation: 1. The project's scope is developed conservatively. Management: 1. The scope is reduced appropriately.
5.	The national lockdown extends to span the entirety of the project timeline.	1	3	Mitigation: 1. The project is structured in such a manner as to be viable even if conducted entirely during lockdown. Management: 1. The project proceeds as normal throughout the lockdown period.
6.	One or more team members are unable to complete their assigned tasks.	3	1	Mitigation: 1. Team members are assigned tasks that can be transferred to other team members easily. 2. Assigned tasks are distributed in such a way as to not overwhelm team members. 3. Tasks are, where deemed necessary, allocated to multiple researchers. Management: 1. The tasks in question are reassigned to other team members.
7.	A member of the *Milk Matters* staff is, for any reason, unable to contribute to the project in any meaningful capacity	2	1	Mitigation: 1. *All Milk Matters* staff are consulted early and frequently about the project, in order to encourage participation. Management: 1. The project progresses, relying wholly on information gleaned from donor mothers and the available staff member.
8.	Mothers are ultimately unhappy with the application and opt not to use it.	2	2	Mitigation: 1. Donor mothers are consulted early and frequently about the project. 2. Donor mothers are asked to evaluate prototypes.
				Management: 1. *Milk Matters* ultimately decides whether to continue hosting the application.
9.	Mothers are driven by the application to donate more milk than they can spare.	4	2	Mitigation: 1. The application is designed from the outset to implement responsible persuasive design. Management: 1. Any features driving this unhealthy persuasion are disabled.

Risk Table from a 2020 Honours Project
Objective: Develop, test, and deploy front-end and backend for a milk bank donor app (Milk Matters)

Some Tips on Project Control

Remember to update planning documents

- Show progress in Gantt Chart or Kanban Board
- Reassess risks and respond regularly



Suggested Reading

Sommerville, *Software Engineering*, Pearson (10e)

- Chapter 22: Project management: Risk management, Managing people, Teamwork
- Chapter 23: Project planning, Software pricing, Plan-driven development, Project scheduling, Agile planning, Estimation techniques